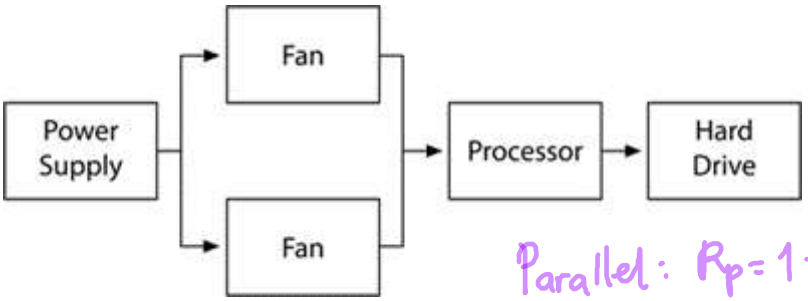
Mechanical Design MS4092

fivan1039@gmail.com (not shared) Switch accounts

Mechanical Design MS4092

Number of problems: 8
Start at: 08.00 PM
Finish at: 08.24 PM
Duration: 24 Minutes

A Reliability Block Diagram (RBD) of a simplified computer system with a redundant fan configuration is shown below. If the reliability of each component is 0.8, what is the system reliability?



Parallel: $R_p = 1 - [(1-0,8)(1-0,8)]$

$R_p = 0,96$

Total: $R_T = 0,8 \times 0,96 \times 0,8 \times 0,8$

$= 0,49$

- ○ 0.72
- ○ 0.41
- ○ 0.81
- ⊗ 0.49

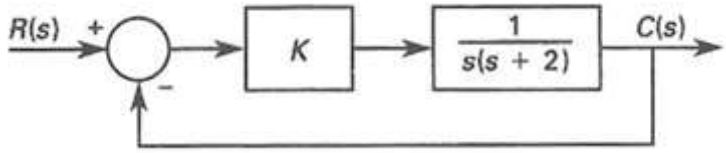
Which of the following failure theories is most suitable for engine components using aluminum?

first aluminium: ductile
↳ shear

- [x] Maximum Shear Stress Theory
- [ ] Modified Mohr theory
- [ ] Maximum Normal Stress Theory
- [ ] Internal Friction Theory

For the given block diagram, K is set to 16. The damped natural frequency of oscillation in response to a step input is most nearly...

step input: $F(s) = 1$



- [x] 3.9
- [ ] 5.9
- [ ] 4.9
- [ ] 6.9

$$\frac{C}{R} = \frac{\frac{K}{s(s+2)}}{1+\frac{K}{s(s+2)}} = \frac{K}{s^2+2s+K}$$

$$\frac{C}{R} = \frac{16}{s^2+2s+16}$$

$$s^2+2\zeta\omega_n s+\omega_n^2$$

$$\omega_n = 4$$

$$2\zeta\omega_n = 2$$

$$2 \cdot \zeta \cdot 4 = 2$$

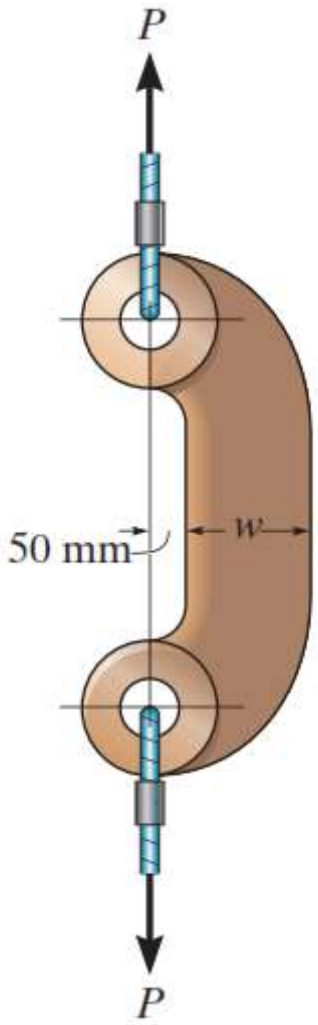
$$\zeta = 0.25$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$= 4\sqrt{1-0.25^2}$$

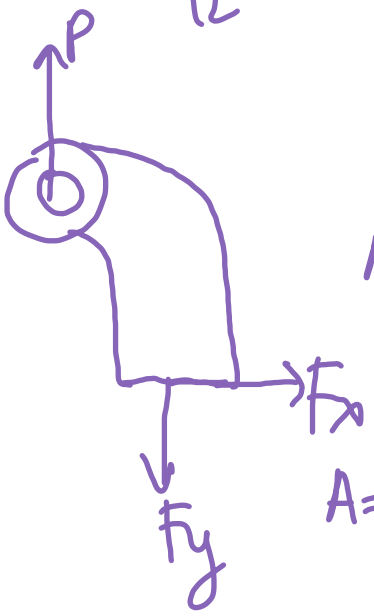
$$= 3.9$$

The offset link supports the loading of P = 30 kN and has a width of w = 200 mm and a thickness of 40 mm. Determine the maximum shear stress of the offset link.



- 16.88 MPa
- 10.32 MPa
- 20.63 MPa
- 8.44 MPa

$I = \frac{1}{12}bh^3 = \frac{1}{12} \cdot 40 \cdot 200^3$



$F_y = 30\ kN$

$M = 4500$

$A = 200 \times 40 = 8 \times 10^3\ mm^2$

$F_y \cdot 150 = F_x \cdot 25$

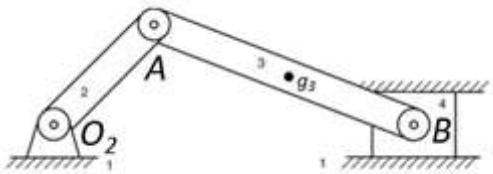
$F_x = \frac{30 \times 150^6}{25} = 180\ kN$

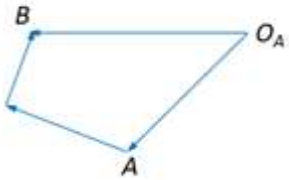
$\tau_1 = \frac{180 \times 10^3\ N}{8 \times 10^3 \times 10^{-6}\ m^2} \cdot \frac{1}{10^6}\ MPa$

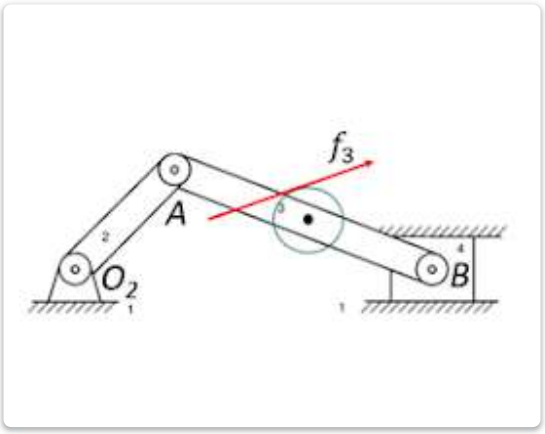
$= 22.5\ MPa$

$\tau = \frac{4500 \times 0.1}{\frac{\pi}{64}(0.2^4 - 0.12^4)} =$

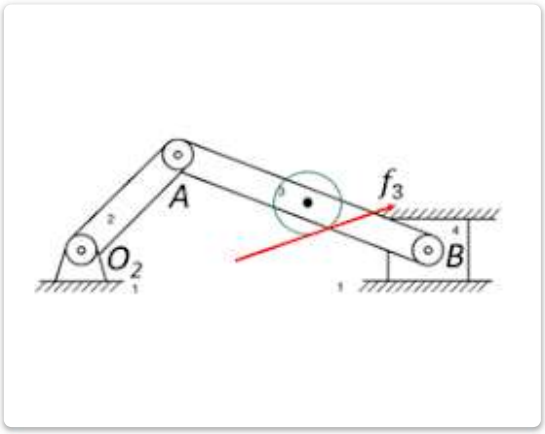
In the slide-crank mechanism and its acceleration polygon below, the correct inertia force of link 3 is...



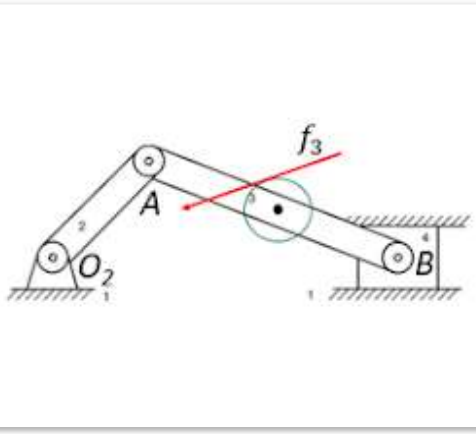




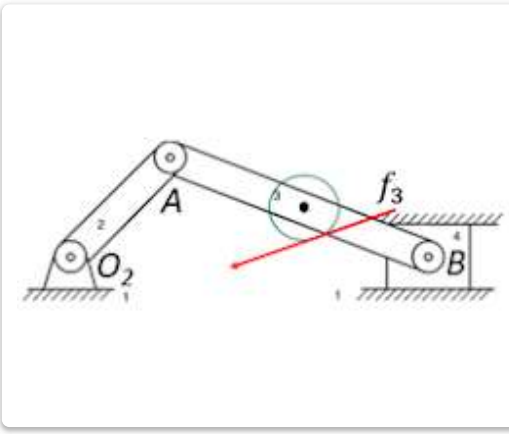
Option 4



Option 2



Option 1



Option 3

The rotor of a steam turbine is rotating at 7200 rpm when the steam supply is suddenly cut off. The rotor decelerates at a constant rate and comes to rest after 5 min. What is most nearly the angular deceleration of the rotor in rad/s^2?

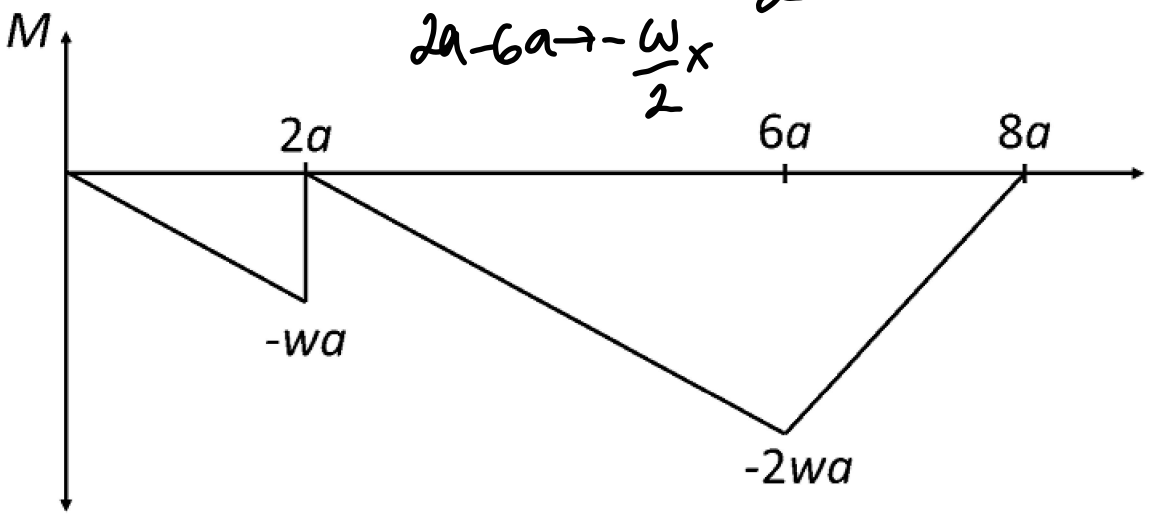
- ○ 0.4
- ⊗ 2.5
- ○ 5.8
- ○ 16

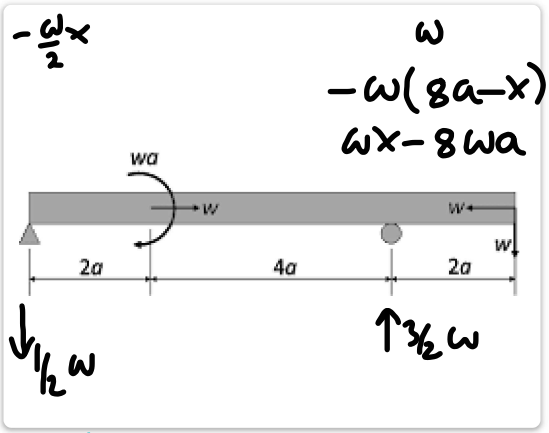
$\alpha = \frac{7200 \text{ rpm}}{5 \text{ min}} = 1440 \text{ rev/min}^2$

$= \frac{1440 \times 2\pi}{60 \times 60}$

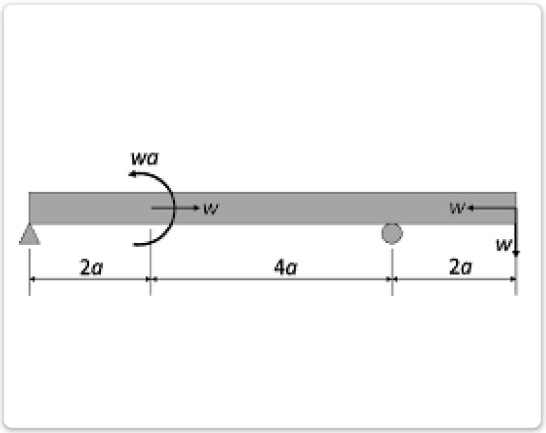
$= 2.5$

Based on the diagram of bending moment below, which loading of a bar is correct?

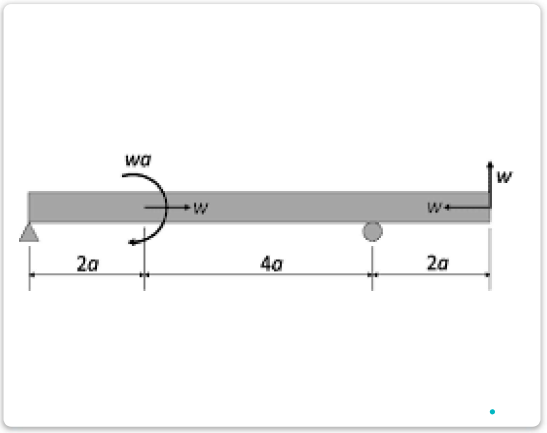




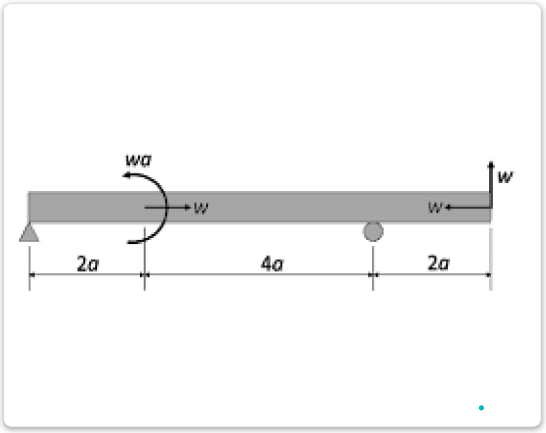
Loading 2



Loading 3



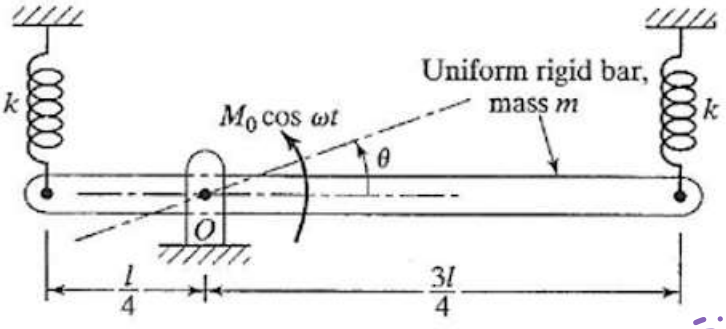
Loading 1



Loading 4

Please answer the following problem.

The equation of motion of the system shown in the figure below for rotational motion about the hinge O is $A\ddot{\theta} + B\theta = C\cos(Dt)$. It is known that $k = 5000$ N/m, $l = 1$ m, $m = 10$ kg, $M_0 = 100$ Nm, $\omega = 100$ rad/s, and $J_o = \frac{7}{48}ml^2$. The value of A, B, C, and D of motion equation are…



- [x] A = 7, B = 15000, C = 480, D = 100
- [ ] A = 7, B = 1200, C = 480, D = 100
- [ ] A = 7, B = 1500, C = 480, D = 100
- [ ] A = 7, B = 12000, C = 480, D = 100

$\Sigma\tau = J_o\ddot{\theta}$

$-k\cdot\frac{3l}{4}\sin\theta\frac{3l}{4} - k\cdot\frac{l}{4}\sin\theta\cdot\frac{l}{4} + M_0\cos\omega t = J_o\ddot{\theta}$

$J_o\ddot{\theta} + \frac{kl^2}{16}\theta + k\cdot\frac{9l^2}{16}\theta = M_0\cos\omega t$

$J_o\ddot{\theta} + \frac{10kl^2}{16}\theta = M_0\cos\omega t$

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$\frac{7}{48}\cdot 10\cdot 1^2\ddot{\theta} + \frac{10}{16}\cdot 5000\cdot 1^2\cdot\theta = 100\cos 100t$

$7\ddot{\theta} + 1500\theta = 480\cos 100t$

$A = 7$

$B = 1500$

$C = 480$

$D = 100$